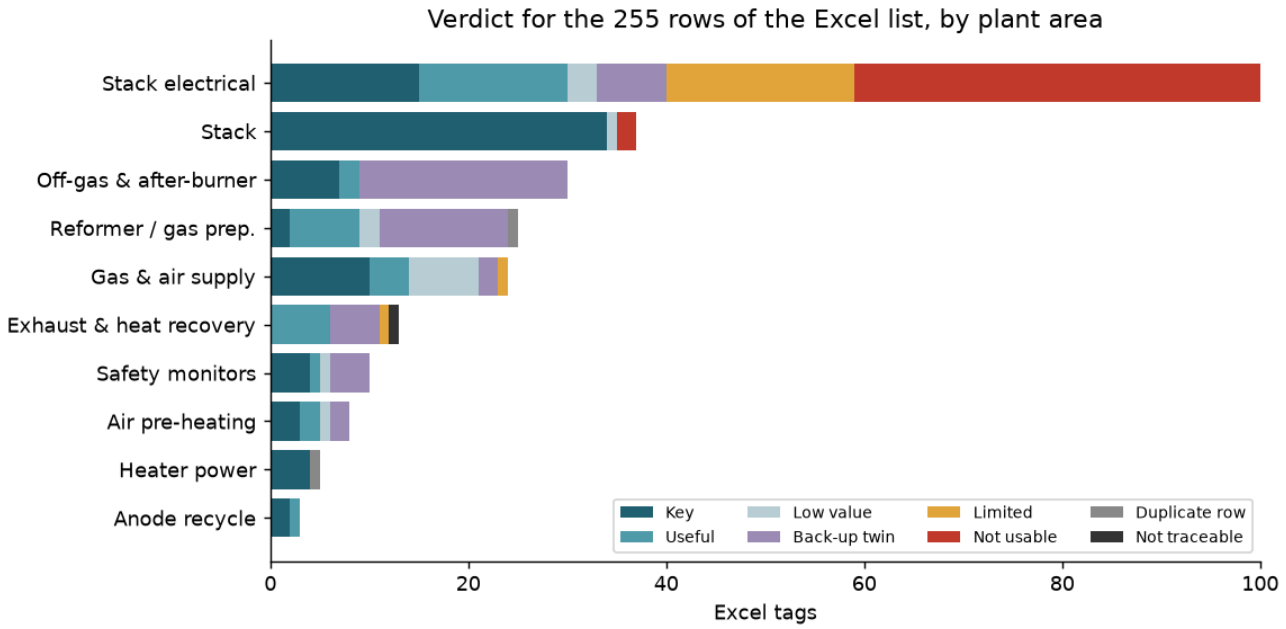
Almost every sensor on the Excel list is in the data; about a third of them clearly carry the plant state

Contents: 1 Overview · 2 Excel name → raw name · 3 Data quality · 4 Correlation. Source: data_analysis/excel_sensor_analysis.ipynb.

The list has 255 rows. 254 can be found in the raw data under a known name (only LIR 7,01 is not logged). 43 channels are dead or wrong, 54 are near-copies of another sensor, 81 carry the plant state clearly. The raw data has 38 measured columns that are missing from the list.

Why it matters: the stage separation and the fake-data generator use these sensors. Dead channels add noise and copies add nothing new.



Area	Key	Useful	Low value	Back-up twin	Limited	Not usable	Duplicate row	Not traceable
Air pre-heating	3	2	1	2	0	0	0	0
Anode recycle	2	1	0	0	0	0	0	0
Exhaust & heat recovery	0	6	0	5	1	0	0	1
Gas & air supply	10	4	7	2	1	0	0	0
Heater power	4	0	0	0	0	0	1	0
Off-gas & after-burner	7	2	0	21	0	0	0	0
Reformer / gas prep.	2	7	2	13	0	0	1	0
Safety monitors	4	1	1	4	0	0	0	0
Stack	34	0	1	0	0	2	0	0
Stack electrical	15	15	3	7	19	41	0	0

Verdict rules in order: not found → duplicate → not usable → limited → back-up twin ($|r| \geq 0.98$ over the whole period and in hot operation) → key (used by the stage rules, a known limit, or the stage explains ≥ 50 % of its variation) → useful (20–50 %) → low value (< 20 %).

If you remember one thing: For stage separation and synthetic data, one sensor per group of near-copies is enough; the 40 near-zero voltage channels and two broken stack thermocouples should be fixed at the source or dropped.

2 · EXCEL NAME → RAW NAME

Most tags keep their name in the raw files; flows, loads and heaters use controller names

Raw files sofc-data/raw/<day>.txt have one header line, 836 column names over 39 days. 201 rows: same name with commas → underscores; 35 documented other name; 18 best match; 1 missing. Full table: data_analysis/excel_sensors/excel_to_raw_names.csv.

Excel tag	Raw column	Match / note
TIR 4,6,1	TIR_4_6_1	direct: commas become underscores
FIRC 1,1,1	ct_FIRC_111_PV_ProcessValue	direct: flow process value
EIRC 4,1,2	PLC_hostSetp_ctrl_ctEIRC41_rCurrMonitor	other name: load of string 1, current
EIRC 4,1,7	EIRC_41_OUT_Enable	other name: load on/off, only from 2026-03-17
EIR 4,1,12	pi_EIR_4_1_12	direct: stack voltage (12/15/18 = stack 1/2/3)
EIRC 2,01	EIRC_2_01_ActV_In_ThreePhasePower	other name: air pre-heater power
EIRC 3,1,1	pi_IIR_3_1_1_ActVal	assumed: heater current
TIRC 3,19,1	TIR_3_19_1	other name: logged as TIR
QIR 1,4	PLC_pi_QIR_1_4_ActVal_Value	other name: natural-gas quality
FIRC 1,1,3	ct_FIRC_113_PV_ProcessValue	only from 2026-03-17; before: PLC_ctrl_ct_FIRC_113_PV_ProcessValue (≈ 0.15)
LIR 7,01	–	not in any raw file

Measured in the raw data but missing from the Excel list

What	Raw columns	Note
Stack-4 voltage of each string	pi_EIR_4_1_21 … pi_EIR_4_5_21	≈ 100 V hot; the Excel list stops at stack 3
Unknown channel per string	pi_EIR_4_s_22	≈ 230–996, not documented
Room climate sensors	pi_QIR_10_23/24/25 Temp, RH, Press	temperature, humidity, pressure ≈ 990–1013 hPa: possible true ambient reference
Load monitor twins (from 03-17)	EIRC_4s_CurrentMon, VoltageMon, ActV_Min_CellVoltage	same quantities as the PLC host monitors
Heater currents	pi_IIR_5_1_3_ActVal, pi_IIR_6_1_1_ActVal	
Others	TIR_2_02, PLC_pv_QIR_1_4_calc	

If you remember one thing: Use the name table when loading data; two useful groups (stack-4 voltages, room climate sensors) are not on the list.

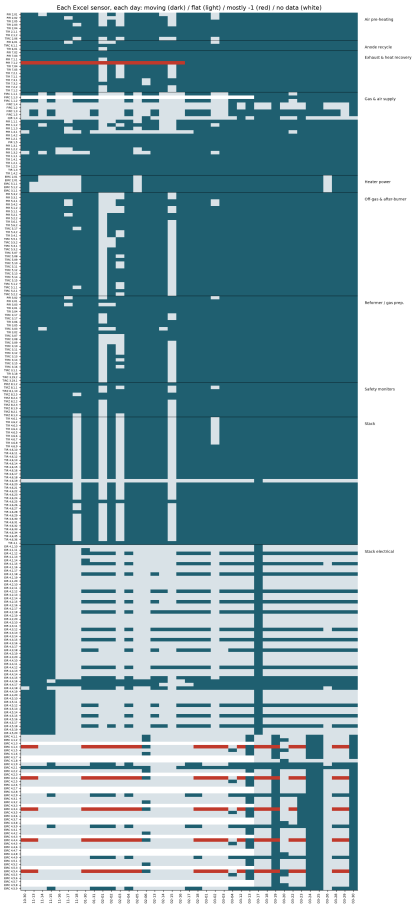
3 · DATA QUALITY

Nearly every channel is logged every day; the problems sit in a few groups

238 of 254 traced tags have values on about 100 % of logged minutes. Flat values on days the plant is off are correct, not faults.

Problem	Tags	What the data shows	Verdict
Near-zero voltages	EIR 4,s,10/11/13/14/16/17/19/20 (40)	≈ 0.005 V; real ≈ 5 V only on 2025-11-14/15	Not usable
Broken thermocouple	TIR 4,6,19	stuck at 1300 °C	Not usable
Misplaced thermocouple	TIR 4,6,25	≈ 36 °C while the stack reads 400–600 °C	Not usable
Dead status	EIRC 4,1,8	never changes	Not usable
Late columns	FIRC 1,1,3, EIRC 4,s,3/7/8	only from 2026-03-17 (12 days)	Limited
Set-points -1 when idle	EIRC 4,s,4	-1 on 56–58 % of minutes	Limited
Connected late	PIR 7,1,2	-1 until 2026-02-17	Limited
Logger dropouts	nearly all	-1 on ≈ 0.1 % of minutes (02-18, 03-13, 03-17); blank them	–

Not a fault: natural gas FIRC 1,4, POx air FIRC 1,2 and the load current first move in February/March, when the plant first used them.

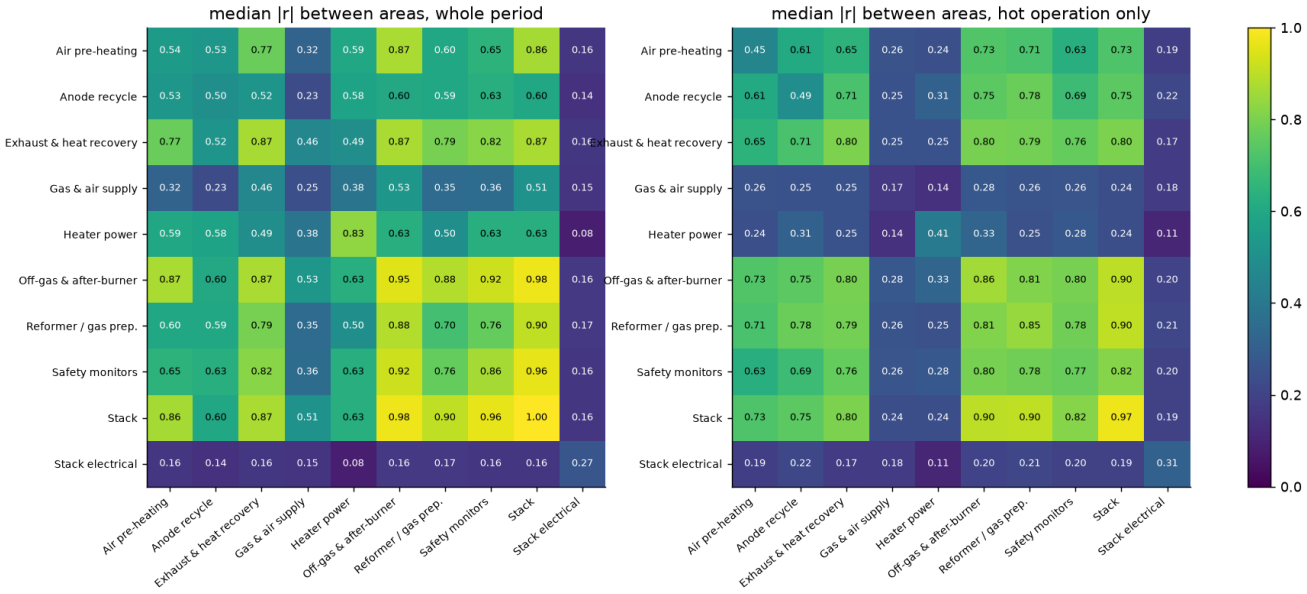


If you remember one thing: Blank the -1 dropout minutes everywhere and treat 43 channels as missing.

4 · CORRELATION

The plant heats and cools as one body; only about 56 groups carry their own information

Spearman r: +1 = rise and fall together, –1 = opposite, 0 = unrelated. Over all days 107 of 209 usable sensors follow the stack temperature with |r| ≥ 0.8; in hot operation (Hot standby, Thermal adjustment, Part load) the median |r| drops from 0.50 to 0.34, but 88 still follow it. Grouping pairs with |r| ≥ 0.95 leaves 56 groups (75 in hot operation).



Near-copies even in hot operation: reformer → off-gas → exhaust temperatures (r ≈ 0.997), control/safety pairs such as TIR 7,05 ↔ TIRZ 8,1,10 (0.995), anode-recycle and off-gas pressures (0.995), POx air ↔ natural gas (0.99).

Finding to check: in hot operation the stack voltages of strings 4 and 5 fall as the stack warms (r –0.45 to –0.83; also EIR 4,2,15 –0.80), strings 1 and 3 hardly react (–0.10 to +0.18). This fits the internal-leakage note from IKTS; the split by string is worth raising with them.



If you remember one thing: Whole-period correlation mostly measures “plant hot or cold”; judge links inside hot operation.